Braided-Symmetric Derivations of Algebras

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ABSTRACT
We study braided-symmetric derivations of an algebra over a field. These are maps that satisfy the Leibniz rule for the product symmetrized by an involution of $A \otimes A$. The choice of involution decides which derivations appear. The identity involution gives the ordinary derivations when 2 is invertible. The swap gives the Jordan derivations. We then twist the swap by an involutive automorphism or anti-automorphism. The Leibniz defect of a map decides when it is a twisted derivation. For a commutative reduced algebra with 2 invertible, the twisted maps that commute with the automorphism are exactly the derivations that commute with it. On the algebra $\mathbb{F}_3[\varepsilon]/(\varepsilon^3)$ the twisted derivations and the α-derivations do not contain each other. An example with nilpotent elements shows that the reduced hypothesis is needed, and an example in characteristic 2 shows that the invertibility of 2 is needed.

KEYWORDS
Braided-symmetric derivations; twisted Jordan derivations; Jordan derivations; Leibniz defect

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1. Introduction

Throughout, K is a field and an algebra means a unital associative K-algebra. We take all maps between algebras to be K-linear. We say so when a result cited below holds more generally for additive maps on rings.

A *derivation* of an algebra A is a linear map $D\colon A \to A$ with $D(xy) = D(x)y + xD(y)$ for all $x, y \in A$. The notion goes back to Jacobson [12, Part I], who studied the derivations of an arbitrary nonassociative algebra. Hochschild [9, §§2 and 4] defined the cocycles of an associative algebra, and its 1-cocycles are the derivations. A *Lie derivation* is a linear map with $D([x, y]) = [D(x), y] + [x, D(y)]$ [14, p. 183]. Benkovič [2, §1] treats derivations, Jordan derivations and Lie derivations as f-derivations, for $f = x_1x_2$, $x_1 \circ x_2$ and $[x_1, x_2]$, so that one identity for a multilinear polynomial f covers all three. Beidar et al. [1, Theorem 4.3] studied maps that act as derivations on a fixed multilinear polynomial, earlier than Benkovič. A *Jordan derivation* of A is a linear map with $D(x \circ y) = D(x) \circ y + x \circ D(y)$ for all $x, y \in A$, where $x \circ y = xy + yx$. The sum $xy + yx$ is twice the Jordan product $\frac{1}{2}(xy + yx)$ of an associative algebra

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[4, Example 2.1]. Jordan, von Neumann and Wigner form a product ab from the two products $a \times b$ and $b \times a$ of an associative system, and call this quasi-multiplication [13, p. 64]. Herstein [7, p. 1104] takes the sum $ab + ba$ without the half. We follow Herstein and do not divide by 2, so that characteristic 2 is allowed. His result needs three notions for a ring R. We call R *prime* if $aRb = 0$ implies $a = 0$ or $b = 0$ [3, p. 1003]. We call R *semiprime* if $aRa = 0$ implies $a = 0$ [3, p. 1003]. We call R 2*-torsion free* if $2a = 0$ implies $a = 0$ [3, p. 1003]. So every prime ring is semiprime. Herstein proved that every Jordan derivation of a prime ring of characteristic different from 2 is a derivation [7, Theorem 3.1]. Every derivation is a Jordan derivation, and the converse is false in general [3, p. 1003]. Cusack [6, Corollary 5] extended this result of Herstein to 2-torsion free semiprime rings, and Brešar [3, Theorem 1] gave an independent proof.

Let A be an algebra over K with product μ. Let $\sigma\colon A \otimes A \to A \otimes A$ be a K-linear map with $\sigma^2 = \mathrm{id}$, for instance the swap $\tau(x \otimes y) = y \otimes x$. It gives a symmetrized product, as the Jordan product does [4, Example 2.1]

$$x \cdot_\sigma y = \mu(x \otimes y) + \mu(\sigma(x \otimes y)).$$

For $\sigma = \tau$ this is the Jordan product $x \circ y$. Put $\mu_\sigma = \mu + \mu \circ \sigma$. Since $\sigma^2 = \mathrm{id}$, we get $\mu_\sigma \circ \sigma = \mu_\sigma$, that is, the following triangle commutes.

$$\begin{array}{ccc} A \otimes A & \xrightarrow{\ \sigma\ } & A \otimes A \\ & {\scriptstyle \mu_\sigma} \searrow & \downarrow {\scriptstyle \mu_\sigma} \\ & & A \end{array}$$

Example 2.3 shows that a twisted version of this product can fail to be associative and to have a unit.

We call a linear map $D\colon A \to A$ a *braided-symmetric derivation* for σ if

$$D(x \cdot_\sigma y) = D(x) \cdot_\sigma y + x \cdot_\sigma D(y). \tag{1}$$

So (1) is the Leibniz rule for the symmetrized product. Other authors have studied *braided derivations*, which use a braiding operator directly [15, §2] [10, §4.1, Definition 14]. Remark 2.2 compares the two identities. Our identity uses only μ, σ and D.

We now describe the main results. Our first step is a dictionary. Remark 2.4 shows that the identity gives exactly the derivations when 2 is invertible, and the swap gives exactly the Jordan derivations.

Our second step twists the swap by an involutive automorphism or anti-automorphism α of A. For an involutive α we put $\sigma_\alpha(x \otimes y) = \alpha(y) \otimes \alpha(x)$, and we write $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$ for the maps of the class that commute with α; we call them *twisted derivations*. Proposition 3.2 describes the twisted class through the defect $E_D(x, y) = D(xy) - D(x)y - xD(y)$. Corollary 3.5 shows that, on 2-torsion-free semiprime rings, the Jordan maps in the twisted class are the derivations commuting with α.

Theorem 4.1 treats commutative reduced algebras and needs no Jordan hypothesis. Example 4.2 shows that the reduced hypothesis is needed, and Example 4.3 shows the same for the invertibility of 2. Proposition 4.4 shows that the twisted class and the α-derivations on one algebra do not contain each other.

We follow this order. Section 2 recalls the classical notions, defines braided-symmetric derivations and treats the identity and the swap. Section 3 treats the twisted

swap through the Leibniz defect. Section 4 treats reduced commutative algebras and compares the twisted class with α-derivations. We place each example right after the result it illustrates.

2. Preliminaries and braided-symmetric derivations

This section recalls the classical notions we need, defines braided-symmetric derivations and identifies them for the identity and the swap.

From now on, A is a K-algebra with unit 1. We write $[x, y] = xy - yx$ for the commutator, $x \circ y = xy + yx$ for the Jordan product, and $Z(A)$ for the centre. The centre is the set of elements z with $zx = xz$ for all $x \in A$. We call an element x *nilpotent* if $x^n = 0$ for some $n \geq 1$. A commutative algebra is *reduced* if its only nilpotent element is zero [5, 1.4.12].

A derivation of an algebra is as defined by Jacobson [12, Part I]: a linear map D with $D(xy) = D(x)y + xD(y)$. Every derivation satisfies $D(1) = 0$, since $D(1) = D(1 \cdot 1) = 2D(1)$. For $g \in A$, the map $\mathrm{ad}_g(x) = [g, x]$ is an *inner derivation*, called a principal derivation by Weibel [16, §9.2]. We write $\mathrm{Der}(A)$ and $\mathrm{Inn}(A)$ for these two spaces.

We add a word on the characteristic for Jordan derivations. A *Jordan derivation* in the sense of Herstein [7, p. 1104] is an additive map with $D(x^2) = D(x)x + xD(x)$. We write $\mathrm{JDer}(A)$ for the space of linear maps with $D(x \circ y) = D(x) \circ y + x \circ D(y)$ for all x, y. Replacing x by $x + y$ gives the condition $D(x \circ y) = D(x) \circ y + x \circ D(y)$ in every characteristic. Setting $y = x$ in this condition gives $2D(x^2) = 2(D(x)x + xD(x))$. So the two conditions agree when 2 is invertible in K, and we use the $\circ$ form throughout. When $\mathrm{char}\, K = 2$ and A is commutative, $x \circ y = 0$, and every linear map satisfies the $\circ$ condition.

For an automorphism α of A, an *α-derivation* satisfies $D(xy) = D(x)\alpha(y) + \alpha(x)D(y)$ [11, §1]. We write $\mathrm{Der}_\alpha(A)$ for this space.

We now compare the Jordan derivations with the derivations. Every derivation is a Jordan derivation [3, p. 1003], so $\mathrm{Der}(A) \subseteq \mathrm{JDer}(A)$.

Definition 2.1. Let A be a K-algebra and let σ be a K-linear involution of $A \otimes A$. A *braided-symmetric derivation* for σ is a K-linear map $D\colon A \to A$ that satisfies identity (1), that is, $D(x \cdot_\sigma y) = D(x) \cdot_\sigma y + x \cdot_\sigma D(y)$ for all $x, y \in A$. We write $\mathrm{BSDer}_\sigma(A)$ for the set of braided-symmetric derivations for σ. This set is a K-subspace of $\mathrm{End}_K(A)$, since the condition is linear in D. For an automorphism or an anti-automorphism α of A, we write $\mathrm{BSDer}_\sigma(A)^\alpha$ for the subset of maps D with $D\alpha = \alpha D$.

Identity (1) says that the following square commutes.

$$\begin{array}{ccc} A \otimes A & \xrightarrow{\mu_\sigma} & A \\ {\scriptstyle D\otimes 1 + 1 \otimes D}\downarrow & & \downarrow{\scriptstyle D} \\ A \otimes A & \xrightarrow[\mu_\sigma]{} & A \end{array}$$

The name is close to an older one, and the next remark separates the two.

Remark 2.2. Braided derivations [15, §2] [10, §4.1, Definition 14] form a different notion from braided-symmetric derivations, although both use a braiding. Definition 2.1 asks instead for the ordinary Leibniz rule for the new product $\mu + \mu \circ \sigma$.

Example 2.3 shows that the product $\cdot_\sigma$ can fail to be associative and unital, even when A is commutative.

Example 2.3. Let $A = K \times K$ with componentwise operations, and let $\alpha(a, b) = (b, a)$. Then α is an automorphism with $\alpha^2 = \mathrm{id}$, and $\sigma_\alpha(x \otimes y) = \alpha(y) \otimes \alpha(x)$ is an involution of $A \otimes A$. Since A is commutative, $x \cdot_{\sigma_\alpha} y = xy + \alpha(xy)$. For $x = (x_1, x_2)$ and $y = (y_1, y_2)$ this gives $x \cdot_{\sigma_\alpha} y = (x_1y_1 + x_2y_2)(1, 1)$. Put $p = (1, 0)$ and $q = (0, 1)$. Then $(p \cdot_{\sigma_\alpha} p) \cdot_{\sigma_\alpha} q = (1, 1) \cdot_{\sigma_\alpha} q = (1, 1)$, while $p \cdot_{\sigma_\alpha} (p \cdot_{\sigma_\alpha} q) = p \cdot_{\sigma_\alpha} 0 = 0$. So $\cdot_{\sigma_\alpha}$ is not associative. Every product $u \cdot_{\sigma_\alpha} y$ lies in $K(1, 1)$, so $u \cdot_{\sigma_\alpha} p \neq p$ for every u, and $\cdot_{\sigma_\alpha}$ has no unit.

The next remark records what Definition 2.1 says for the involutions that use no structure of A.

Remark 2.4. Let A be a K-algebra and let σ be a K-linear involution of $A \otimes A$. Identity (1) for σ is the Leibniz rule for the product $\cdot_\sigma$. So $\mathrm{BSDer}_\sigma(A)$ is the set of derivations of the nonassociative algebra $(A, \cdot_\sigma)$, in the sense of Jacobson [12, Part I].

For the identity, $x \cdot_{\mathrm{id}} y = 2xy$, so the identity reads

$$2D(xy) = 2\big(D(x)y + xD(y)\big).$$

If $\operatorname{char} K \neq 2$, we divide by 2 and get $\mathrm{BSDer}_{\mathrm{id}}(A) = \mathrm{Der}(A)$. If $\operatorname{char} K = 2$, then $x \cdot_{\mathrm{id}} y = 0$, so $\mathrm{BSDer}_{\mathrm{id}}(A)$ contains every linear map. So it is larger than $\mathrm{Der}(A)$. For example, take the linear map with $D(1) = 1$ and $D = 0$ on a complement of $K1$. Then D is not a derivation, since a derivation satisfies $D(1) = D(1 \cdot 1) = 2D(1) = 0$ in characteristic 2.

For the swap τ we get $x \cdot_\tau y = xy + yx = x \circ y$. So the defining identity is the $\circ$ form of the identity for a Jordan derivation recalled above. Hence $\mathrm{BSDer}_\tau(A) = \mathrm{JDer}(A)$ in every characteristic.

Every derivation is a Jordan derivation [3, p. 1003]. So in every characteristic

$$\mathrm{Der}(A) \subseteq \mathrm{BSDer}_\tau(A) = \mathrm{JDer}(A).$$

When $\operatorname{char} K \neq 2$, this gives $\mathrm{BSDer}_{\mathrm{id}}(A) \subseteq \mathrm{BSDer}_\tau(A)$.

The involutions $-\mathrm{id}$ and $-\tau$ use no structure of A either, and they give the two remaining entries of this dictionary. Their products are

$$x \cdot_{-\mathrm{id}} y = xy - xy = 0, \qquad x \cdot_{-\tau} y = xy - yx = [x, y].$$

So $\mathrm{BSDer}_{-\mathrm{id}}(A) = \mathrm{End}_K(A)$, since every linear map is a derivation of the zero product, and $\mathrm{BSDer}_{-\tau}(A)$ is the space of Lie derivations of A [14, p. 183]. In characteristic 2 the four involutions collapse to two, $-\mathrm{id} = \mathrm{id}$ and $-\tau = \tau$.

3. Twisted involutions and the Leibniz defect

Remark 2.4 treated the identity and the swap, which use no structure of A. This section twists the swap by an involution α of A and treats it with one tool, the Leibniz defect.

We recall the notions used here. An *automorphism* of A is a bijective K-linear map $\alpha\colon A \to A$ with $\alpha(xy) = \alpha(x)\alpha(y)$ and $\alpha(1) = 1$. An *anti-automorphism* of A

is a bijective K-linear map $\alpha\colon A \to A$ with $\alpha(xy) = \alpha(y)\alpha(x)$ and $\alpha(1) = 1$. By $(xy)^T = y^T x^T$, the transpose is an anti-automorphism of $M_n(K)$. In both cases α is *involutive* if $\alpha^2 = \mathrm{id}$. An involutive anti-automorphism is an involution of the ring A in the sense of Herstein [8, §1.2], that is, a map $*$ with $a^{**} = a$, $(a+b)^* = a^* + b^*$ and $(ab)^* = b^* a^*$.

We need $\alpha^2 = \mathrm{id}$ for the following reason. Applying σ_α below twice gives

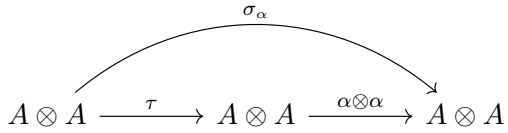
$$x \otimes y \mapsto \alpha^2(x) \otimes \alpha^2(y),$$

so σ_α is an involution of $A \otimes A$ exactly when α is involutive. When 2 is invertible, an involutive α splits $A = A_+ \oplus A_-$, where $A_\pm = \{x \in A : \alpha(x) = \pm x\}$. Indeed, for $x \in A$,

$$x = \tfrac{1}{2}\big(x + \alpha(x)\big) + \tfrac{1}{2}\big(x - \alpha(x)\big).$$

We call the elements of A_+ *symmetric* and the elements of A_- *skew*.

Let α be involutive, of either kind. Then $\sigma_\alpha(x \otimes y) = \alpha(y) \otimes \alpha(x)$ is a K-linear involution of $A \otimes A$. It is the composite of the swap and $\alpha \otimes \alpha$, as the following diagram shows.



So σ_α defines a symmetrized product as in Section 2. The product it defines is

$$x \cdot_{\sigma_\alpha} y = xy + \alpha(y)\alpha(x). \tag{2}$$

For $\alpha = \mathrm{id}$ the product μ_{σ_α} is the Jordan product. If α is an automorphism, then $\alpha(y)\alpha(x) = \alpha(yx)$. If α is an anti-automorphism, then $\alpha(y)\alpha(x) = \alpha(xy)$.

Definition 3.1. The *Leibniz defect* of a linear map $D\colon A \to A$ is the bilinear map $E_D\colon A \times A \to A$ given, for $x, y \in A$, by

$$E_D(x, y) = D(xy) - D(x)y - xD(y).$$

The name is ours, and the map is classical: Cusack [6, §2] writes it $d(a, b) = D(ab) - aDb - (Da)b$ and studies Jordan derivations through it. We recall the Hochschild complex of A with coefficients in A [16, §9.1]. An *n-cochain* is a K-multilinear map from A^n to A. So a 0-cochain is an element of A, and a 1-cochain is a linear map $D\colon A \to A$. The *coboundary* δ sends each n-cochain to an $(n+1)$-cochain, and $\delta \circ \delta = 0$ [16, §9.1]. An *n-cocycle* is an n-cochain f with $\delta f = 0$. The n-th *Hochschild cohomology* $HH^n(A, A)$ is the space of n-cocycles modulo the images δg of the $(n-1)$-cochains g [16, §9.1].

On a 1-cochain D the coboundary is $(\delta D)(x, y) = xD(y) - D(xy) + D(x)y$ [9, §2, Definition 2.1]. So $E_D = -\delta D$. Thus D is a derivation exactly when $E_D = 0$, so derivations are the 1-cocycles. Since $\delta \circ \delta = 0$, the defect is a 2-cocycle: for all $a, b, c \in A$,

$$aE_D(b, c) - E_D(ab, c) + E_D(a, bc) - E_D(a, b)c = 0. \tag{3}$$

This says that E_D is a cocycle of the Hochschild complex [16, §9.1].

The defect also describes the other classes of Section 2. Expanding $D(x \circ y)$ shows that D is a Jordan derivation exactly when $E_D(x,y) + E_D(y,x) = 0$ for all x, y.

Proposition 3.2. *Let A be a unital associative K-algebra, let α be an involutive automorphism or anti-automorphism of A, and let $D\colon A \to A$ be a linear map with $D\alpha = \alpha D$.*

(i) If α is an automorphism, then $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)$ if and only if $E_D(x,y) + \alpha(E_D(y,x)) = 0$ for all $x, y \in A$.

(ii) If α is an anti-automorphism, then $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)$ if and only if $E_D(x,y) + \alpha(E_D(x,y)) = 0$ for all $x, y \in A$.

In both cases every derivation that commutes with α lies in $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$.

Proof. Write $E = E_D$. By Remark 2.4, $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)$ means $D(x \cdot_{\sigma_\alpha} y) = D(x) \cdot_{\sigma_\alpha} y + x \cdot_{\sigma_\alpha} D(y)$ for all x, y. We compute the difference of the two sides in each case.

For (i), the product (2) is $x \cdot_{\sigma_\alpha} y = xy + \alpha(yx)$. Since $D\alpha = \alpha D$, the left side is $D(xy) + \alpha(D(yx))$. The right side is $D(x)y + \alpha(yD(x)) + xD(y) + \alpha(D(y)x)$. We group the terms without α and the terms with α. The first group gives $D(xy) - D(x)y - xD(y) = E(x,y)$. Since α is linear, the second group gives $\alpha\big(D(yx) - D(y)x - yD(x)\big) = \alpha(E(y,x))$. So the left side minus the right side is $E(x,y) + \alpha(E(y,x))$, which proves (i).

The proof of (ii) follows the same steps as the proof of (i). Here the product (2) is $x \cdot_{\sigma_\alpha} y = xy + \alpha(xy)$. Since $D\alpha = \alpha D$, the left side is

$$D(xy) + \alpha(D(xy)),$$

and the right side is

$$D(x)y + \alpha(D(x)y) + xD(y) + \alpha(xD(y)).$$

The same grouping gives the difference

$$E(x,y) + \alpha(E(x,y)),$$

which proves (ii).

If D is a derivation, then $E = 0$, and both conditions hold. □

We now read the two conditions, assuming that 2 is invertible.

In case (ii), the equation $E_D + \alpha(E_D) = 0$ says $\alpha(E_D) = -E_D$. So E_D takes values in A_-, the skew elements of α. In case (i), we write $E_D = E_\mathrm{s} + E_\mathrm{a}$. Here $E_\mathrm{s}(x,y) = \frac{1}{2}(E_D(x,y) + E_D(y,x))$ is symmetric. And $E_\mathrm{a}(x,y) = \frac{1}{2}(E_D(x,y) - E_D(y,x))$ is antisymmetric.

We swap x and y in the condition of (i). Then we take the sum and the difference of the two equations. So the condition holds exactly when E_s takes values in A_- and E_a takes values in A_+. Now let $\alpha = \mathrm{id}$. Then $A_- = 0$ and $A_+ = A$, so the condition says $E_\mathrm{s} = 0$. This is the Jordan condition, so Proposition 3.2(i) recovers the swap case of Remark 2.4.

Remark 3.3. Let α be an involutive automorphism of A, and let $D \in \mathrm{Der}(A)$. Then

$E_D = 0$. By (2), the two sides of the σ_α-identity differ by

$$D(\alpha(yx)) - \alpha(D(yx)).$$

Since A is unital, the products yx span A. So $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)$ if and only if $D\alpha = \alpha D$.

Next we look at the inner derivations ad_g in this class. The condition $\mathrm{ad}_g\,\alpha = \alpha\,\mathrm{ad}_g$ means

$$[g - \alpha(g), \alpha(x)] = 0 \qquad \text{for all } x.$$

Since α is bijective, this holds if and only if $g - \alpha(g) \in Z(A)$. Let $\mathrm{Inn}_\alpha(A)$ be the space of the maps ad_g with $g - \alpha(g) \in Z(A)$. These are the inner derivations in $\mathrm{BSDer}_{\sigma_\alpha}(A)$, so $\mathrm{Inn}_\alpha(A) = \mathrm{Inn}(A) \cap \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. The quotient $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha / \mathrm{Inn}_\alpha(A)$ is the space of *twisted outer derivations*.

Cusack [6, Corollary 5] and Brešar [3, Theorem 1] proved that a Jordan derivation of a 2-torsion free semiprime ring is a derivation.

Theorem 3.4 ([3, Theorem 1])**.** *Let R be a 2-torsion free semiprime ring. Then every Jordan derivation $D\colon R \to R$ is a derivation.*

The next corollary applies Theorem 3.4 to the twisted class.

Corollary 3.5. *Let A be a 2-torsion-free semiprime K-algebra with an involutive K-algebra automorphism α. Then $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha \cap \mathrm{JDer}(A) = \mathrm{Der}(A)^\alpha$, where $\mathrm{Der}(A)^\alpha$ is the set of derivations commuting with α (not to be confused with the space $\mathrm{Der}_\alpha(A)$ of α-derivations of Section 2).*

Proof. Every derivation is a Jordan derivation by Remark 2.4. With Proposition 3.2, this gives $\mathrm{Der}(A)^\alpha \subseteq \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha \cap \mathrm{JDer}(A)$. Conversely, every Jordan derivation of A is a derivation by Theorem 3.4, which applies since a K-linear map is additive; the $\circ$ form of the Jordan condition and Herstein's x^2 form agree by the linearization of Section 2, which uses the 2-torsion-freeness assumed here. So a map in the intersection is a derivation that commutes with α. □

4. Reduced commutative algebras and twisted derivations

For commutative algebras the Jordan hypothesis in Corollary 3.5 can be dropped. Recall from Section 2 that a reduced commutative algebra has no nonzero nilpotent element.

Theorem 4.1. *Let A be a commutative reduced K-algebra with 2 invertible in K, and let α be an involutive automorphism of A. Then $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha = \mathrm{Der}(A)^\alpha$.*

Proof. Every derivation commuting with α lies in $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$ by Proposition 3.2. Conversely, let $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. Write $E = E_D$. Since A is commutative, E is symmetric. So E takes values in A_- by Proposition 3.2(i).

Since D commutes with α, it maps each of A_+ and A_- into itself. So if x and y lie in the same part, then $E(x, y)$ lies in

$$A_+ \cap A_- = 0.$$

Now let $x \in A_+$ and $y \in A_-$, and put $z = E(x, y)$. Then z lies in A_-, and so do xy, zx and xz, while zy lies in A_+. Apply the cocycle identity (3) to $(a, b, c) = (z, x, y)$:

$$zE(x, y) - E(zx, y) + E(z, xy) - E(z, x)y = 0.$$

The pairs (zx, y) and (z, xy) lie in A_-, so E vanishes on them, and we get $z^2 = E(z, x)y$. Apply (3) to $(a, b, c) = (x, z, y)$:

$$xE(z, y) - E(xz, y) + E(x, zy) - E(x, z)y = 0.$$

The pairs (z, y) and (xz, y) lie in A_-, and the pair (x, zy) lies in A_+, so E vanishes on all three and $E(x, z)y = 0$. Since E is symmetric, $z^2 = E(z, x)y = E(x, z)y = 0$. So $z = 0$ because A is reduced. Since E is bilinear and symmetric and $A = A_+ \oplus A_-$, we get $E = 0$. Hence D is a derivation. The proof uses the commutativity of A only to make E_D symmetric, and reducedness only to pass from $z \in A_-$ and $z^2 = 0$ to $z = 0$. It therefore gives the stronger statement in which "reduced" is replaced by "no nonzero $z \in A_-$ satisfies $z^2 = 0$", a condition that is empty when $\alpha = \mathrm{id}$. □

Example 4.2 shows that the reduced hypothesis cannot be dropped, and Example 4.3 shows the same for the invertibility of 2.

Example 4.2. Let $A = \mathbb{F}_5[u, v]/(u^2, v^2)$. It has basis $\{1, u, v, uv\}$. Let α fix u and send v to $-v$. Let D send uv to uv and the other three basis elements to 0. Then D commutes with α. It satisfies the Leibniz rule for $x \cdot_{\sigma_\alpha} y$ on all basis pairs. So $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. But D is not a derivation, since

$$D(uv) = uv \qquad \text{while} \qquad D(u)v + uD(v) = 0.$$

A direct computation on basis pairs, which suffices since both sides of each identity are linear in D and bilinear in the pair, verifies all three statements. Here $z = E(u, v) = uv$ is a nonzero element with $z^2 = 0$.

The next example shows that the invertibility of 2 is needed in Theorem 4.1.

Example 4.3. Let $A = \mathbb{F}_2 \times \mathbb{F}_2$, which is commutative and reduced, and let α swap the two factors. Put $p = (1, 0)$, $q = (0, 1)$ and $e = (1, 1)$, and let D be the linear map with $D(p) = D(q) = e$, so that $D(e) = 0$. Then D commutes with α. On the basis pairs (p, p), (q, q) and (p, q), both sides of the Leibniz rule for $x \cdot_{\sigma_\alpha} y = xy + \alpha(xy)$ are zero. Here $p \cdot_{\sigma_\alpha} p = e$, so the two sides vanish because $D(e) = 0$ and $2e = 0$, not because the products vanish. So $D \in \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. But D is not a derivation: since $p^2 = p$, a derivation would satisfy $D(p) = D(p^2) = 2pD(p) = 0$, while $D(p) = e$.

Our second natural candidate for the twisted class is the space $\mathrm{Der}_\alpha(A)$ of α-derivations. It was defined for automorphisms in Section 2. It differs from $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$ in both directions, and one algebra shows both failures.

Proposition 4.4. *Let $A = \mathbb{F}_3[\varepsilon]/(\varepsilon^3)$, and let α be the automorphism with $\alpha(\varepsilon) = -\varepsilon$. Then*

$$\mathrm{Der}_\alpha(A) \not\subseteq \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha \qquad \textit{and} \qquad \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha \not\subseteq \mathrm{Der}_\alpha(A).$$

Proof. The algebra A has basis $\{1, \varepsilon, \varepsilon^2\}$. The map α fixes 1 and ε^2 and sends ε to $-\varepsilon$. So it is an involutive automorphism with $A_+ = \mathbb{F}_3 1 + \mathbb{F}_3 \varepsilon^2$ and $A_- = \mathbb{F}_3 \varepsilon$. Since A is commutative, E_D is symmetric for every linear map D. Let D be a linear map with $D\alpha = \alpha D$. By Proposition 3.2(i), D lies in $\mathrm{BSDer}_{\sigma_\alpha}(A)$ exactly when E_D takes values in A_-. We give one map for each non-inclusion. Both maps are diagonal on the basis, so both commute with α.

For the first non-inclusion, let $D(a + b\varepsilon + c\varepsilon^2) = b\varepsilon + c\varepsilon^2$. We check that D is an α-derivation on pairs of basis elements. Pairs that contain 1 give $D(y)$ on both sides, since $D(1) = 0$ and $\alpha(1) = 1$. For the pair $(\varepsilon, \varepsilon)$, the left side is $D(\varepsilon^2) = \varepsilon^2$. The right side is $D(\varepsilon)\alpha(\varepsilon) + \alpha(\varepsilon)D(\varepsilon) = -2\varepsilon^2 = \varepsilon^2$. The remaining pairs have product 0, and each term on the right side is a multiple of $\varepsilon^3 = 0$. So $D \in \mathrm{Der}_\alpha(A)$. On the other hand, $E_D(\varepsilon, \varepsilon) = D(\varepsilon^2) - 2\varepsilon D(\varepsilon) = \varepsilon^2 - 2\varepsilon^2 = 2\varepsilon^2$. This element is nonzero and lies in A_+, so it does not lie in A_-. Hence $D \notin \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$.

The second non-inclusion uses a derivation.

Let d be the linear map

$$d(a + b\varepsilon + c\varepsilon^2) = b\varepsilon + 2c\varepsilon^2.$$

It is a derivation, since $d(\varepsilon^i) = i\varepsilon^i$ and exponents add under multiplication. So $E_d = 0$, and $d \in \mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$. On the other hand,

$$d(\varepsilon^2) - d(\varepsilon)\alpha(\varepsilon) - \alpha(\varepsilon)d(\varepsilon) = 2\varepsilon^2 + 2\varepsilon^2 = \varepsilon^2 \neq 0.$$

Hence $d \notin \mathrm{Der}_\alpha(A)$. □

The two classes can also meet; the next example shows a case where $\mathrm{Der}_\alpha(A)$ is undefined and a case where the two classes coincide.

Example 4.5. Let $A = M_2(\mathbb{F}_3)$ and let α be the transpose, an anti-automorphism. Then $\mathrm{Der}_\alpha(A)$ is not defined, since it needs an automorphism. Let g be skew for the transpose and ad_g the inner derivation $x \mapsto [g, x]$. It commutes with α and has $E = 0$, so it lies in $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$ by Proposition 3.2.

Now let $A = K[\varepsilon]/(\varepsilon^2)$ with $\alpha(\varepsilon) = -\varepsilon$. The map $D(a + b\varepsilon) = b\varepsilon$ is a derivation and an α-derivation, and it commutes with α. So it lies in both $\mathrm{Der}_\alpha(A)$ and $\mathrm{BSDer}_{\sigma_\alpha}(A)^\alpha$ by Proposition 3.2.

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